

Clean Air Act: Statutory Overview

This overview states only facts confirmed against EPA documentation (sources listed at the end). It describes the statutory architecture of the Clean Air Act (CAA); for the current status of specific rules and standards, consult current EPA sources.

The statute

The Clean Air Act is codified at 42 U.S.C. §7401 et seq. EPA’s summary notes the Act “was amended in 1977 and 1990”; the NAAQS program reflects amendments through 1990.

National Ambient Air Quality Standards (NAAQS)

The Clean Air Act requires EPA to set National Ambient Air Quality Standards (40 CFR part 50) for six principal (“criteria”) air pollutants:

- carbon monoxide, lead, nitrogen dioxide, ozone, particulate matter, and sulfur dioxide.

EPA sets two kinds of standards:

- **Primary** standards “provide public health protection, including protecting the health of ‘sensitive’ populations such as asthmatics, children, and the elderly.”
- **Secondary** standards “provide public welfare protection, including protection against decreased visibility and damage to animals, crops, vegetation, and buildings.”

NAAQS concern ambient air quality in a geographic area rather than any individual source.

State Implementation Plans (SIPs)

Under Section 110 of the Clean Air Act, each state must adopt and submit to EPA for approval a SIP providing for the maintenance, implementation, and enforcement of the NAAQS, including a permit program for the construction and modification of stationary sources. SIP requirements are federally enforceable under Section 113 of the Act (reference 40 CFR Part 52). Where a state does not produce an adequate plan, a Federal Implementation Plan (FIP) may be used.

Hazardous air pollutants (Section 112)

Section 112 of the Clean Air Act addresses emissions of hazardous air pollutants (HAPs) — pollutants associated with serious health effects, including cancer. EPA currently lists 188 hazardous air pollutants. Section 112 directs EPA to set technology- and performance-based standards — “maximum achievable control technology” (MACT) standards, issued as National Emission Standards for Hazardous Air Pollutants (NESHAP). MACT standards are promulgated under 40 CFR Part 63.

Source and permitting programs

The following CAA programs are recorded as regulatory air programs in EPA’s facility systems:

- **Prevention of Significant Deterioration (PSD).** Part C of the Act sets requirements to prevent significant deterioration of air quality in areas designated attainment or unclassifiable, and requires a major emitting facility to install best available control technology (per Section 165).
- **Title V operating permits.** Under 40 CFR Part 70, the operating-permit program requires all major stationary sources (and certain other sources) to hold a comprehensive operating permit; Title V does not impose new requirements but provides a single permit assuring compliance with applicable requirements, with a duration not to exceed five years.
- **Acid Rain Program.** Requires major reductions of sulfur dioxide and nitrogen oxide emissions from electric utilities through a “cap and trade” approach, with continuous emissions monitoring of affected sources.
- **New Source Review (NSR) and New Source Performance Standards (NSPS).** Recorded as CAA air programs applicable to stationary sources.
- **Federally Enforceable State Operating Permit (FESOP).** Establishes federally enforceable limits on a source’s potential to emit, keeping it below the Title V applicability threshold (used for sources classified synthetic minor).
- **Stratospheric ozone protection (Title VI).** EPA programs protecting the ozone layer, including CFC tracking (reference 40 CFR Part 82).

Sources

- EPA, *Summary of the Clean Air Act*, <https://www.epa.gov/laws-regulations/summary-clean-air-act>
- EPA, *Criteria Air Pollutants*, <https://www.epa.gov/criteria-air-pollutants>
- EPA, *NAAQS Table*, <https://www.epa.gov/criteria-air-pollutants/naaqs-table>
- EPA, *What are Hazardous Air Pollutants?*, <https://www.epa.gov/haps/what-are-hazardous-air-pollutant>
- EPA, *AFS Data Download* documentation (April 2014) — program descriptions for SIP, PSD, NESHAP/MACT, Title V, Acid Rain, FESOP, and CFC tracking.